

Final Project Report

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DS-UA 202: Responsible Data Science

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Background

We chose the Kaggle "Home Credit Default Risk" competition because it touches on basically everything we've talked about this semester — fairness across demographic subgroups, the use of alternative data in high-stakes decisions, and auditing models that are otherwise pretty opaque. We picked Will Koehrsen's notebook "Start Here: A Gentle Introduction" as our ADS to audit. It's the most upvoted public solution to the competition and scores 0.756 on the public leaderboard, so it felt like a reasonable baseline to dig into.

The ADS itself is a machine learning model that predicts whether a loan applicant will repay (Target = 0) or struggle to make payments (Target = 1). What makes it more interesting than a typical credit-scoring problem is Home Credit's stated mission: they want to lend to "unbanked" people who don't have traditional credit histories, using alternative data sources to fill in the gaps. So there's a real tension built into the goal. If the model leans too strict, it ends up rejecting exactly the people Home Credit claims it wants to serve, which kind of defeats the point. If it leans too generous, the company loses money on defaults and borrowers end up stuck with debt they can't repay.

The competition uses ROC-AUC as its scoring metric, which we think is actually a reasonable choice — it measures ranking quality without forcing you to pick a threshold up front. But as we found later in our audit, that just punts the threshold decision (and the fairness consequences that come with it) to whoever ends up deploying the model.

Input & Output

The data is provided by [Homecredit](#), an international service with the purpose of offering loans to the unbanked population. It consists of real-world loan application records collected from Home Credit's own operational systems and supplemented with information from external sources such as credit bureaus. Because the data reflects only applicants who actually applied for a Home Credit loan, it represents a self-selected subpopulation rather than the general population, which is an important limitation we revisit in the Outcomes and Summary sections. The data provided by Homecredit contains 9 separate files, including one main training file, one main testing file, and 6 supplemental files containing historical credit and payment records from other institutions. In this specific baseline ADS, it only used the `application_train.csv` and `application_test.csv` to train the machine learning model. The two application train and test data include demographic information (e.g., `CODE_GENDER`, `DAYS_BIRTH`), financial

information (e.g., **AMT_INCOME_TOTAL**, **AMT_CREDIT**), and alternative data (normalized scores from external sources, like telco and transactional info) which is stated by ADS creator that they have the strongest mathematical relationship with whether the person defaults or not based on correlation analysis, and helps assess the creditworthiness of "unbanked" populations who lack traditional credit histories.

For the input feature, the main dataset contains 122 features, excluding the **SK_ID_CURR** (the unique ID for the loan application) and **TARGET** (the label we are trying to predict: 1 for default, 0 for repaid), other 120 columns are actuarial predictor features(input features we considered) consisting of 16 categorical features and 105 numerical features. We summarized feature types in aggregate and detail the most predictive variables below (selected based on the correlation analysis in Figure 3):

- **EXT_SOURCE_1, EXT_SOURCE_2, EXT_SOURCE_3** (numerical, float $\in [0, 1]$): Normalized risk scores from three external data sources. Missingness: 56.4%, 0.2%, and 19.8% respectively. These are the three most negatively correlated features with **TARGET** (correlations: -0.16 , -0.16 , -0.18), indicating that higher external scores predict successful repayment.
- **DAYS_BIRTH** (numerical, integer): Client's age in days at the time of application. No missing values. Range corresponds to roughly 20–70 years (Figure 2).
- **CODE_GENDER**: Approximately 66% Female, 34% Male, and 4 records labeled 'XNA'. Used as a sensitive attribute in our fairness analysis.
- **AMT_INCOME_TOTAL** (numerical, float): Self-reported total annual income. Contains extreme outliers (max $> 100M$), which is why we use quartile-based binning (Q1–Q4) rather than fixed thresholds in our income subgroup analysis.

The dataset has a large amount of missing data: 67 columns contain missing values, with several features missing over 60% of their values. Notably, The high missingness in **EXT_SOURCE_1** (detailed in the feature list) directly reflects Home Credit's focus on serving 'unbanked' individuals who lack traditional credit histories. We highlight the distributions of two key variables. Firstly, the **TARGET** variable is heavily imbalanced: around 92% of loans were repaid (Target = 0) and only 8% defaulted (Target = 1) . Secondly, client age (**DAYS_BIRTH**) follows a roughly uniform distribution across adult ages, as shown in Figure 1.

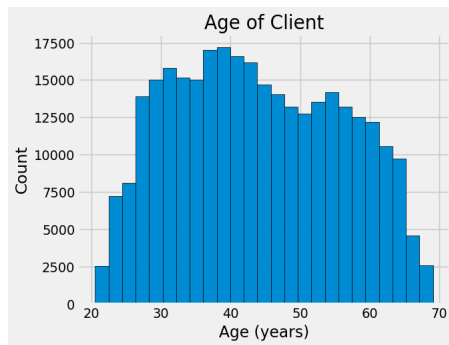


Figure 1

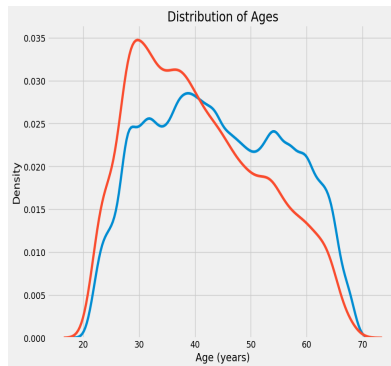


Figure 2

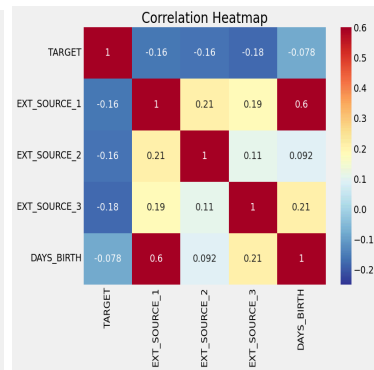


Figure 3

As further profiling of the input data, we analyzed the relationship between the input features and the **TARGET** variable. While most features had very weak linear correlations (Figure 2), the **DAYS_BIRTH** is the most positive correlation and target == 1 curve skews towards the younger end of the range that younger clients have higher failure-to-repay rates, a pattern we will revisit in our age-based subgroup audit in Outcome. We can see a clear correlation that younger clients are less likely to repay the loan. The failure to repay rate is above 10% for the three youngest age groups and below 5% for the oldest age group.

For Pairwise correlation we looked at how our key features overlap, specifically age and the three external risk scores (Figure 3). The key overlap is between **EXT_SOURCE_1** and **DAYS_BIRTH** (0.6) and the rest of the features are nearly independent. The correlations between each external source are quite low, and age has almost no relationship with **EXT_SOURCE_2**. It confirms that our top features help the model see the client's risk profile from several different dimensions.

Finally, the output of the system is a probability score ranging between 0 and 1, which represents the likelihood that a client defaults on the loan.

-A probability closer to 1 indicates a high risk that the client will fail to repay the loan.

-A probability closer to 0 indicates a high likelihood that the client will successfully repay the loan on time.

Implementation and Validation

For data cleaning and pre-processing, the author identified hidden abnormal values of 365243 in the **DAYS_EMPLOYED** column (equivalent to ~1000 years of employment, clearly a placeholder for missing data). These values were replaced with NaN, and a new flag column **DAYS_EMPLOYED_ANOM** was created to preserve the information that these entries were originally anomalous. This design allows the model to learn whether "missingness itself" is predictive of default. For the missing values across 67 columns, the author applies median imputation using `SimpleImputer(strategy='median')` after train/validation split to avoid data leakage. No columns are dropped despite some having >60% missingness. The author assumes that the model can learn from the missingness pattern itself. Additionally, the dataset contains 16 categorical features that the author handles with a two encoding strategy: Label Encoding for the

3 columns with 2 or fewer unique values, and One-Hot Encoding by `pd.get_dummies` for variables with more categories expanding the feature space from 122 to 243 columns. Finally, the author applies `MinMaxScaler` to scale all features to the $[0, 1]$ range. This is necessary for Logistic Regression since it is sensitive to feature scale, but has no effect on Random Forest. The pipeline is: raw data $\rightarrow$ anomaly correction $\rightarrow$ categorical encoding $\rightarrow$ train/validation split $\rightarrow$ median imputation $\rightarrow$ MinMax scaling $\rightarrow$ model fitting $\rightarrow$ probability prediction. And the author trains two models: **logistic Regression** with $C=0.0001$ and $\text{max_iter}=1000$, to prevent overfitting on the 240-feature input; **random Forest** with $n_estimators=100$, $\text{random_state}=50$, and unlimited tree depth, using as a non-linear baseline that captures feature interactions Logistic Regression can't. Both output a **probability of default** $\in [0, 1]$ rather than a binary decision, leaving threshold selection to the lender.

The Random Forest's top features (`EXT_SOURCE_2`, `EXT_SOURCE_3`, `DAYS_BIRTH`) align with our correlation analysis in Section Input and output. However, `SK_ID_CURR` (the loan ID), an identifier that should have no predictive power, also appears in the top 10 with importance 0.030, suggesting the model is fitting noise. The author should have dropped this column before training.

The original Kaggle implementation trains on the whole `application_train.csv` and submits predictions on the test set to Kaggle, where they're scored by ROC-AUC. ROC-AUC is a valid choice. It measures how well the model ranks defaulters above non-defaulters rather than depending on a specific cutoff. It also handles the 92:8 class imbalance well, but accuracy would be misleading that a model that simply predicts "no default" for everyone would score around 92%. Because the public test set doesn't include ground-truth labels, we created an 80/20 stratified split from the training data for local validation. Logistic Regression achieved a ROC-AUC of 0.688 and Random Forest 0.706 (Figure 4), which is consistent with the Kaggle baseline and confirms that our reproduction is on the right track.

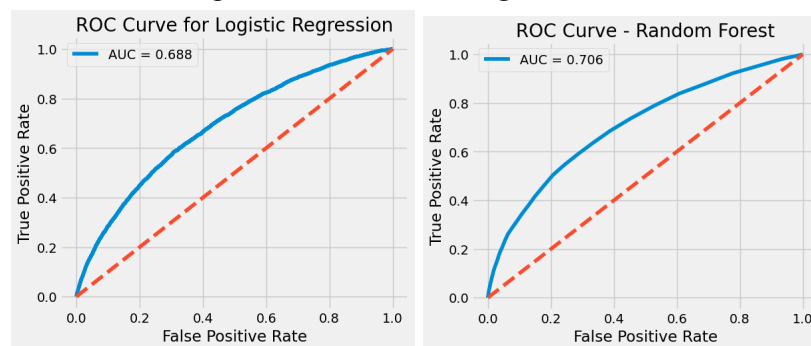


Figure 4

Does this validation establish that the ADS meets its stated goals? We state it does not. The author's validation only establishes that the model can, on average, distinguish defaulters from non-defaulters. It doesn't show that the model is fair across subgroups; it doesn't show that the predicted probabilities are well-calibrated, so we don't know whether a 0.3 prediction really means a 30% chance of default; it doesn't provide interpretability beyond a basic feature

importance ranking; it relies on a single train/validation split rather than cross-validation. These are properties that determine whether the system can be used in the real world.

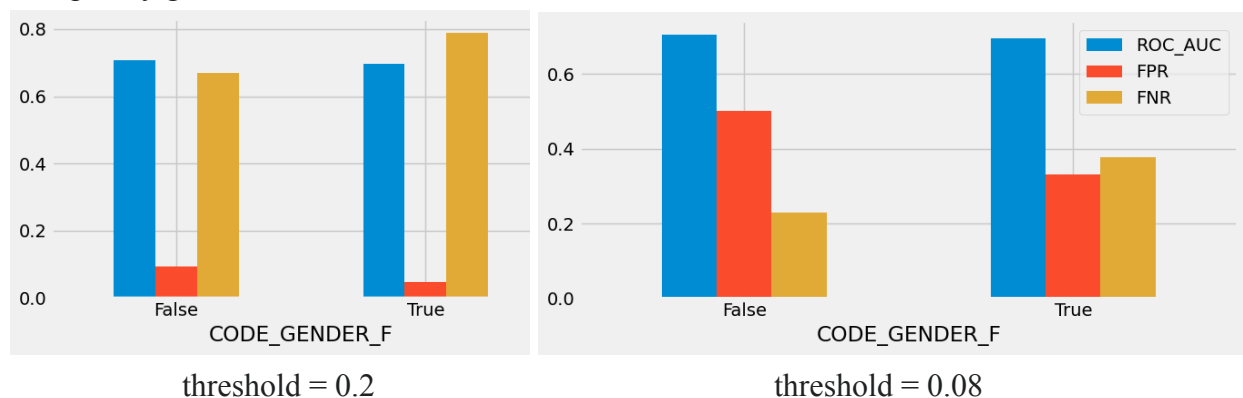
Outcomes

Accuracy Across Subgroups

To measure how the model performs on different subgroups and consider its fairness, we calculated accuracy metrics of the baseline random forest model by dividing the data into several subpopulations, such as gender, age group, and income. For each subpopulation, we calculated predicted default rate, ROC-AUC, accuracy, FPR, and FNR. Since the dataset is highly imbalanced, a threshold of 0.5 (typically used in machine learning settings) may be overly conservative. Therefore, we repeated the process for two more thresholds: 0.2 and the empirical default rate (about 0.08).

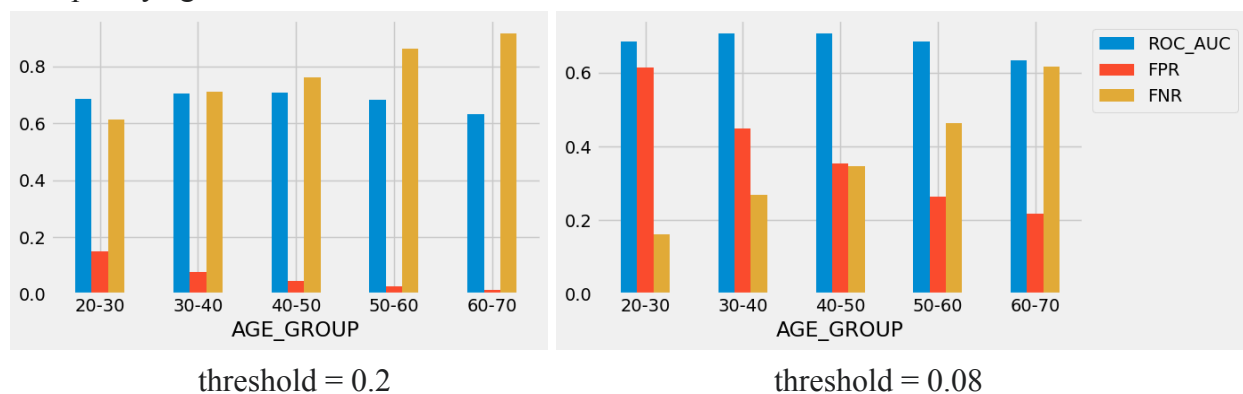
When the threshold is 0.5, the model does not predict any positive (default) result, which is clearly unrealistic. So we decided not to include this part of the result.

Grouped by gender:



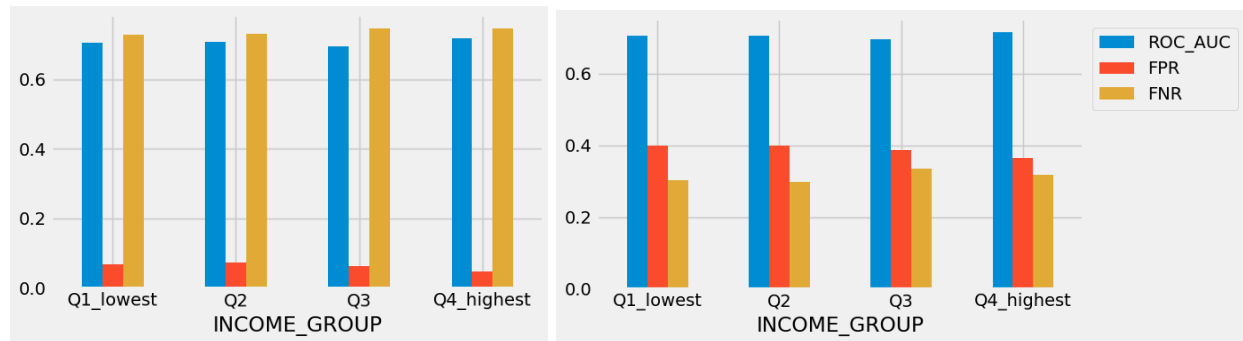
From the graphs we can see that while the ROC-AUC is almost the same, the FPR for female applicants is lower than male applicants, meaning that the model is more likely to reject creditworthy applicants that are male, which is different from what we originally expected.

Grouped by age:



When grouped by age, there is a clear trend that FPR decreases as age increases, and FNR goes the opposite way. The model is more likely to reject younger creditworthy applicants, which is consistent with our previous finding of negative correlation between age and default rate. Also, the ROC-AUC is lower for the elder age group, suggesting that the ranking performance is lower for older applicants.

Grouped by income:



threshold = 0.2

threshold = 0.08

The model achieves relatively similar ROC-AUC values across income groups. The FPR decreases slightly as income increases, which means that the model is less likely to reject trustworthy applicants with higher income.

For all subgroups, the accuracy is higher when threshold = 0.2, the recall is higher when threshold = 0.08, and ROC-AUC is similar between these thresholds. What's more, when the threshold is 0.08, FPR is significantly higher, and FNR is visibly lower compared to the 0.2 threshold case. It is easy to understand that the model with threshold = 0.08 is stricter, and therefore is better at identifying people who fail to pay the loan. It inevitably results in a higher likelihood of rejecting low risk applicants.

Fairness Metrics

We calculated the fairness metrics for gender and age groups using Fairlearn.

Threshold	Demographic Parity Difference	Equal Opportunity Difference	Equalized Odds Difference	Demographic Parity Ratio
0.20	0.058	0.121	0.121	0.507
0.08	0.177	0.150	0.170	0.666

Fairness metrics of different genders

According to the above table, when the threshold is 0.08, the predicted default rate of male and female applicants differs by 17.7%, and their recall differs by 15%. The demographic parity ratio and equalized odds difference also show clear disparities.

Threshold	Demographic Parity Difference	Equal Opportunity Difference	Equalized Odds Difference	Demographic Parity Ratio
0.20	0.162	0.294	0.294	0.087
0.08	0.421	0.393	0.408	0.345

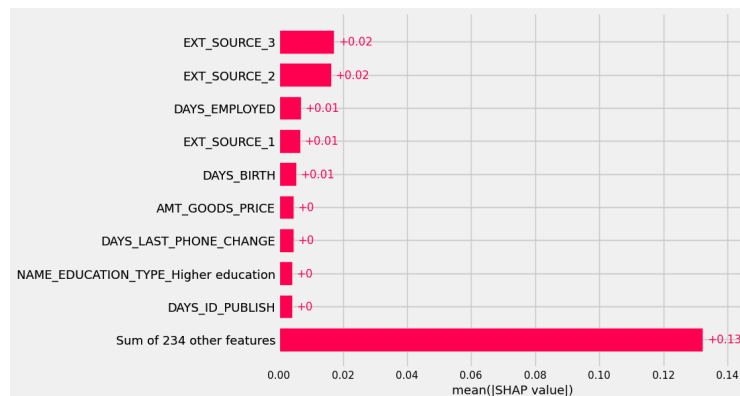
Fairness metrics of different age groups

It seems that fairness disparities across age groups are substantially larger than those observed across gender groups. Demographic parity difference, equal opportunity difference, and equalized odds difference are all higher compared to gender groups. Also, demographic parity ratio when threshold = 0.2 is very low (0.087), showing large differences of predicted positive rates between age groups.

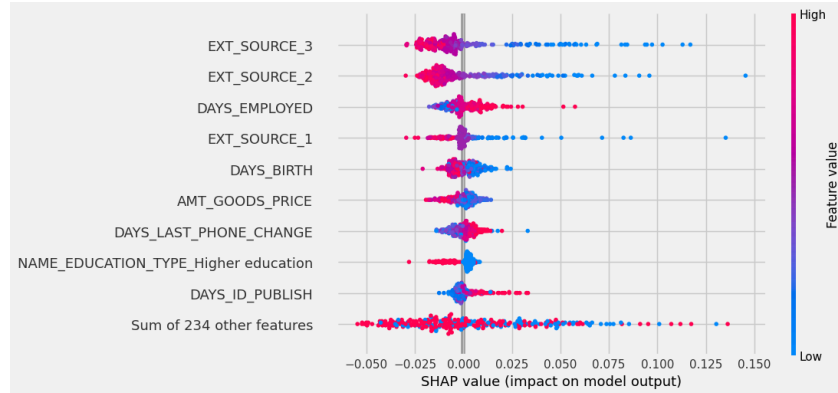
Using a lower threshold increases sensitivity to risky applicants, but also amplifies disparities in prediction rates across demographic groups. However, lowering the threshold increases demographic parity ratios. Maybe this is because predicted positive rates become more similar across groups as it increases.

SHAP Interpretability

While the fairness metrics reveal substantial disparities across demographic groups, they do not explain which features are driving the model's predictions. To better understand the model's decision-making process, we conduct an interpretability audit using SHAP values. Because too many features makes the explainer run slowly, we only sampled 300 applicants from the data.



The SHAP global importance analysis shows that the model relies heavily on the EXT_SOURCE variables, particularly EXT_SOURCE_3 and EXT_SOURCE_2. These features contribute more strongly to the model's predictions than most demographic or financial variables. It may cause applicants without such information ("unbanked" people) to receive unreliable predictions from it. It is also worth noting that although the model depends heavily on EXT_SOURCE, the predictions are also under the cumulative influence of many other features.



For EXT_SOURCE features, the red dots are mainly on the left side, and the blue dots reside on the right side. It means that higher values of EXT_SOURCE features generally push predictions toward lower default risk, while lower values increase predicted risk.

Missingness Audit

Because EXT_SOURCE variables are external and opaque credit-related signals with substantial missingness, applicants without traditional credit histories may receive less reliable predictions after imputation. This raises concerns about whether the system truly supports financial inclusion for “unbanked” applicants. Here we define “Unbanked” as the applicants with any of the three exterior sources missing. We calculate the accuracy metrics separately based on this division. We use the threshold = 0.08 for the following analysis.

Unbanked	n	True default rate	Predicted default rate	ROC_AUC	FPR	FNR
False	21958.0	0.072	0.484	0.718	0.463	0.250
True	39545.0	0.085	0.471	0.702	0.446	0.271

Applicants missing exterior sources (“unbanked”) exhibit slightly worse predictive performance, including lower ROC-AUC and higher false negative rates. The model has a higher likelihood of failing to recognize risky applicants with missing credit history, which demonstrates worse risk detection ability.

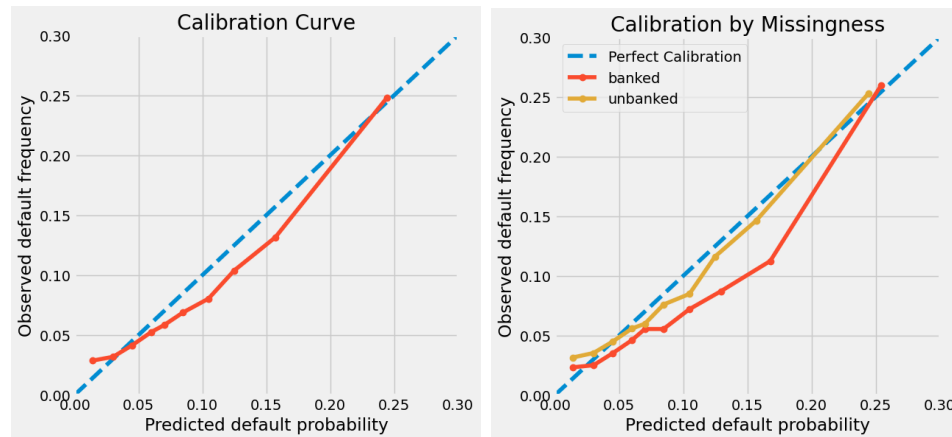
Calibration Analysis

Finally, we move on to analyze the calibration performance of the model, an important indicator of model performance in the risk detection field. Poor calibration may indicate that the model systematically overestimates or underestimates risk for certain groups. Besides the overall calibration curve, we also drew the calibration curve across gender groups and unbanked groups. The diagonal line means perfect calibration (predicted probability = true probability).

The overall calibration curve is mostly under the diagonal line, meaning that the model is conservative: it tends to overestimate the default risk of applicants.

Interestingly, although the unbanked group exhibits slightly worse ROC-AUC and higher false negative rates, its calibration curve remains relatively close to the ideal calibration line. This

suggests that while the model is somewhat less effective at ranking risk within the unbanked population, its predicted probabilities still broadly correspond to observed default frequencies. In contrast, the banked group shows higher overestimation of risk in the intermediate probability range.



As mentioned above, before training models, the author of this ADS imputed all columns with missing values using median imputation, which may induce bias. In particular, we do not know whether the missingness in the dataset is completely random. Some missing values, especially in important variables such as the EXT_SOURCE features, may be related to applicants' financial backgrounds or access to traditional credit systems. As a result, replacing these missing values with median values may reduce important differences between applicants and make predictions for some groups less reliable. This may explain why those applicants show slightly worse performance, such as lower ROC-AUC and higher false negative rates.

Summary

Looking at the dataset overall, we'd say it's only partially appropriate. It's large and real-world, and includes the kinds of features a credit model needs, like demographics, financials, and the external scores supporting Home Credit's "unbanked" mission. But a few things bothered us. The biggest one is that it's a self-selected sample: only people who applied for a Home Credit loan show up, and TARGET is only observed for those who got approved, so the model is really learning from a survivor population. Missingness is also severe that 67 columns have missing values and EXT_SOURCE_1 alone is missing for 56% of applicants. DAYS_BIRTH and CODE_GENDER both correlate with TARGET, meaning "neutral" features end up acting as proxies for protected attributes. And with only an 8% positive rate, threshold choice matters a lot, but the data gives no real guidance.

On performance, our take is that the ADS works reasonably well on average, but with real concerns about fairness and robustness. ROC-AUC stayed relatively stable across groups and the predicted probabilities broadly tracked observed default rates. But once we sliced by subgroup, the picture got messier. Meaningful fairness gaps appeared across demographic groups, especially by age, and they shifted depending on the threshold. Lowering the threshold caught

more defaulters but inflated both false positive rates and fairness disparities. The model also leaned heavily on EXT_SOURCE variables, which are exactly the features most often missing for unbanked applicants, and since the author used median imputation, we suspect those predictions are less reliable than they look.

On metric choices: we used ROC-AUC for overall ranking, and FPR and FNR for the two types of harm. FPR captures the cost to applicants who get unfairly rejected; FNR captures the financial risk to the lender. Demographic parity and equal opportunity matter most for applicants and regulators, while our missingness audit was specifically aimed at "unbanked" applicants who are most likely to lack the features the model relies on.

So would we be comfortable seeing this deployed? Honestly, we'd be pretty cautious. We don't think it should go into industry or public-sector use without a lot more review and ongoing monitoring, especially around the unbanked population and the choice of threshold. If we were prioritizing fixes, we'd start with improving how missing values are handled (probably moving past median imputation) and finding ways to reduce reliance on opaque external credit scores. Overall the ADS performs reasonably well, but after this audit we came away thinking that balancing predictive accuracy, fairness, and financial inclusion is genuinely hard, and probably not something a single model can solve on its own.

Project Contributions

The work for this project was divided between the two team members as follows:

Ashley Ying:

- Wrote the Background section of the report
- Wrote the Input and Output section of the report
- Wrote the Implementation and Validation section of the report
- Wrote the Summary section
- Wrote the Contributions section
- Created the presentation slides
- Overall report revising

Suri Su:

- Drafted the initial Background section
- Implemented the code for Part 4
- Wrote the Part 4 corresponding report
- Wrote the Summary section
- Overall report revising

Works Cited

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